Yu Zhang

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Research Interests

Large Language Models (LLMs), Natural Language Processing (NLP) with Reinforcement Learning (RL), Multimodal NLP, Interdisciplinary Clinical NLP

Education

Cornell University | Ithaca, US *Aug 2025 – Dec 2026 (Expected)*

M.Eng. Systems Engineering | GPA: 3.96/4.0

Core Modules: AI Reasoning & Decision-Making, Clinical NLP & LLMs, Natural Language Processing, Advanced Machine Learning (LLMs)

University of Southampton | Southampton, UK *Sep 2021 – Sep 2022*

M.Sc. Artificial Intelligence | GPA: 3.5/4.0

Core Modules: Intelligent Agents, Artificial Intelligence, Machine Learning, Advanced Machine Learning, Computer Vision, Reinforcement and Online Learning, Deep Learning

Xi'an Jiaotong-Liverpool University &University of Liverpool | Suzhou, China & Liverpool, England *Sep 2016 – Jul 2020*

B.Eng. Computer Science and Electronic Engineering

Core Modules: Database Development, Programming in Java, C++ Programming, Software Engineering

Research Experience

Generative Multimodal Intelligence for Sustainability Audit | Cornell University, US *Sep 2025 – Present*

Researcher (with Dr. Xinlai Liu)

- Developed a structured RAG evidence database from corporate sustainability reports with traceable provenance, quality flags, and context-aware evidence packaging.
- Designed a hybrid evidence-unit construction method combining dense/lexical retrieval, topic expansion, deduplication, and 16-topic evidence weighting.
- Generated SFT-ready evidence pools for 16 GRI disclosure topics, including strong positives, weak evidence, hard negatives, and materiality-noise cases.
- Produced 16,383 evidence units and 4,961 SFT candidate samples from a 100-report calibration across 50 companies for future audit scoring and firm-ranking model training.
- Built credible evidence-pair and SFT data infrastructure for a sustainability-auditing LLM, supporting an EMNLP-style research direction in evidence-linked vertical LLM training.

Machine Learning-based Text Generation | Heilongjiang Artificial Intelligence Institute, China *Nov 2022 – Feb 2023*

Team Leader

- Led a research project analyzing 60+ social media topics (500 posts, 3k comments each) to advance neural text generation
- Designed a modular pipeline combining LDA, SVR and BERT, contributing to a peer-reviewed publication on contextual content synthesis
- Developed an emotion- and topic-aware comment generator with PLSA and ConPRG, improving diversity and contextual relevance

Control of Repetitive Tasks using Reinforcement Learning | University of Southampton, England *Mar 2022 – Sep 2022*

Researcher

- Modeled second-order dynamic systems using iterative learning control (ILC) and norm-optimal ILC in MATLAB
- Integrated Q-table into classical ILC structures and conducted controlled experiments
- Built a neural-network-free Actor-Critic ILC framework and verified its convergence on a simulated printer control system
- Tuned core hyperparameters to benchmark algorithm performance under diverse dynamic conditions

BERTScore Reproduction for Text Generation Evaluation | University of Southampton, England *Mar 2022 – May 2022*

Team Leader

- Rebuilt the BERTScore evaluation pipeline using multilingual WMT18 corpus subsets across five languages
- Applied pre-trained models (BERT, RoBERTa, XLNet, XLM) for cross-lingual machine translation and semantic alignment
- Verified result reproducibility and measured variance against the original paper

Deep Reinforcement Learning for StarCraft II Gameplay | University of Liverpool, England *Sep 2019 – May 2020*

Researcher

- Simulated StarCraft II environments with 6 Protoss structures, 7 weapons and 16 random actions via the official API
- Integrated A3C and PPO to build a distributed PPO network for asynchronous policy updates across 4 parallel environments
- Trained agents over 400 iterations, achieving 73%/51%/39% win rates against simple/medium/advanced AI and validating policy generalization

Professional Experience

BinXian Zeqiu Xiaoquan Fresh Supermarket | China *Sep 2023 – Jul 2025*

Co-Founder & Engineer

- Oversaw 5 supermarket locations with 80K RMB ($15K) daily revenue, 600+ daily customers, and 20K+ SKUs
- Streamlined procurement and supplier onboarding, deployed data-driven membership campaigns, developed dynamic pricing models, and automated inventory tracking to cut discrepancies by 20%
- Managed ERP and database systems, re-built the 20K+ SKU supply chain databases, and rolled out digital membership systems
- Built automated invoicing tools that handled 300K+ RMB monthly, reducing workload by 70%

DataMIMO | Remote, US *Jun 2023 – Aug 2023*

Data Scientist (Volunteer)

- Automated data extraction from Morningstar and Macrotrends using Python
- Designed multithreaded C++ pipelines that lowered data retrieval latency by 24%
- Implemented quantitative optimization and Random Forest models, lifting the Sharpe ratio by 0.1

Suzhou Speedlight| China *Mar 2021 – Jul 2021*

Data Analytics Intern

- Analyzed multi-domain data streams using packet-capture techniques
- Collected large-scale datasets and implemented ad-targeting segmentation
- Developed an adaptive data capture system with compression algorithms

Publications

- **Y. Zhang**, "Path Semantics Enhanced Meta-Learning Framework for Few-Shot Knowledge Graph Reasoning," in Proc. 7th Int. Conf. Advances in Electrical Engineering and Computer Applications (AEECA), Dalian, China, 2026 (accepted).
- **Y. Zhang**, "Incorporating emotional trend into multi-emotion analysis models for long-text sentiment analysis," in Proc. Int. Conf. Machine Learning and Applications (ICMLA), 2023, doi: 10.54254/2755-2721/44/20230241.
- Z. Yang, H. Xie, Y. Xu, Q. Xu, **Y. Zhang**, S. Zhao, H. Yuan, and Y. Fang, "Evaluation of smart response systems for city emergencies and novel UV-oriented solution for integration, resilience, inclusiveness and sustainability," in Proc. 5th Int. Conf. Universal Village (UV), 2020, doi: 10.1109/UV50937.2020.9426215.
- **Y. Zhang**, "Application and development of computer artificial intelligence technology," Journal of Computer Science Research, 2019, ISSN: 2630-5151

Awards & Extracurricular

- MIT Universal Village Summer Research (Team Lead, Emergency System)
- University Entrepreneurship Competition (3rd Prize, "Word Arena" project)
- Bali Marine Conservation Volunteer

Technical Skills

- **Languages**: Mandarin (Native), English (Fluent)
- **Programming**: Python, C++, Java, R, MATLAB
- **Tools**: PyTorch, TensorFlow, Scikit-learn, Keras